Shahan Baksh

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EDUCATION

Wilfrid Laurier University, Waterloo, ON Expected 2028
Bachelor of Computer Science
Relevant Coursework: Data Structures II, Object-Oriented Programming, Discrete Structures, Databases, Software Engineering, Probability & Statistics

TECHNICAL SKILLS

Languages: Go, TypeScript, JavaScript, Python, SQL
Backend: Node.js, Express, PostgreSQL, Redis, REST APIs
Frontend: React, Next.js, React Native, Tailwind CSS, Vite, Expo
Tools: Git, Docker, Fly.io, Anthropic / Claude API, Server-Sent Events

PROJECTS

Distributed Rate Limiter + Analytics Pipeline rate-limiter-shahanb06.fly.dev
Go · Redis · Python · PostgreSQL · Next.js · Docker · Fly.io github.com/shahanb06/rate-limiter

- Built a distributed rate limiter in Go + Redis with fixed-window, sliding-window, and token-bucket algorithms as atomic Lua scripts (consistent under load, no distributed locks); benchmarked at ~5,000 req/s, p95 ~2 ms, and zero failures across ~106K requests (k6).
- Designed an event-sourced analytics pipeline: each decision emits a fire-and-forget event to a Redis Stream, which a Python worker rolls into per-minute PostgreSQL aggregates, keeping reporting off the latency-critical /check path.
- Built a live Next.js analytics dashboard (per-key leaderboard, rejection-rate gauges, per-algorithm latency breakdown) backed by a read-only Go analytics API isolated from the rate-limit hot path.
- Deployed all three services to production on Fly.io and Vercel, with the limiter load-balanced across two machines as a 9.2 MB static Go binary.

Alignr: AI Resume Tailoring Tool alignrai.net
React · TypeScript · Vite · Node.js · Express · Anthropic API github.com/shahanb06/alignr

- Built and deployed a full-stack web app that tailors resumes to job descriptions using a constraint-driven prompt design that prevents fabricated content and flags missing skills honestly.
- Implemented real-time response streaming via Server-Sent Events with a custom client-side parser, handling progress, content, and completion events to replace blocking request UX.
- Designed a strict-JSON API with server-side input validation, per-IP rate limiting, and in-memory parsing of PDF / DOCX / TXT uploads (no disk writes).
- Deployed across Vercel (frontend) and Render (backend) with a custom domain, environment-based builds, and CORS configuration; debugged a production routing bug by tracing request origin in browser DevTools.

Scotia Spark: Investing Onboarding Concept *Scotiabank case competition*
React Native · Expo · TypeScript github.com/shahanb06/ScotiaSpark

- Built the Account and Learn screens for Scotia Spark, a mobile investing-onboarding app prototyped in a 4-person team at [case]Hacks 2026, May 2026, using React Native + Expo and TypeScript.
- Developed the Account dashboard surfacing investable balance across multiple accounts, per-account goal progress (e.g. TFSA targets), and CDIC-protection status, using react-navigation and react-native-svg for data visualization.
- Implemented the Learn module, a bite-sized financial education feed, with expo-haptics for interaction feedback and AsyncStorage for local progress state.
- Contributed to a compliance-by-design reward model that rewards completed lessons and savings milestones rather than trading frequency or login streaks, grounded in TD Bank Group and OSC research on young Canadian investors.

EXPERIENCE

Apollo Health and Beauty Care (via Timely Personnel Ltd.) May 2025 to Aug 2025
IT Technical Support

- Provided technical support for the employee time and attendance system, resolving punch-in / punch-out issues to keep shift records accurate.
- Administered employee access cards, processing batch activations and deactivations to manage facility and system access.
- Supported payroll processing by reviewing and reconciling employee hours against recorded punch data.

LEADERSHIP & EXTRACURRICULAR

The Citizens Foundation, Laurier Chapter (TCF Laurier) Sep 2024 to Present
Vice President of Marketing

- Lead social media and promotional campaigns for the chapter's fundraising drives and member-engagement events, helping raise over $6,000 toward The Citizens Foundation's education programs.